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Search for New Physics in anomalous jets using jet substructures with the ATLAS detector and LHC Run 3 data

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Valerio Tinari
ID number 1998628

Advisor
Prof. Stefano Giagu

Co-Advisors
Dr. Valerio Ippolito
Dr. Giuliano Gustavino

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Sapienza University of Rome

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Author's email: tinari.1998628@studenti.uniroma1.it

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Introduction

It was the 4 July 2012 when Joseph Incandela, CMS spokesperson, affirmed: "*They combine to give us a combined significance of five standard deviations*". With these exact words, he announced, confirmed by ATLAS spokesperson Fabiola Gianotti a few minutes later, the first observation of the Higgs boson [1, 2], which completed the Standard Model (SM) framework. Fourteen years later, no comparable discovery has followed within the SM paradigm at collider experiments, strengthening the hypothesis that it is the most complete and well-tested description of nature.
In reality, SM probably is not as complete as it appears. In fact, there are many phenomena not included in the framework but confirmed by astrophysical and cosmological observations: among these we can find a quantum description of gravity, Dark Matter [3] and baryon asymmetry. All these processes fall in the category of New Physics (NP) Beyond the Standard Model (BSM). They have been hunted by ATLAS and CMS experiments using traditional data analysis techniques, targeting specific theoretical models but, unfortunately, nothing has been observed yet and only upper limits on production cross-sections were set. There are many reasons for this unexpected outcome: wrong Physical model for experiments, wrong triggering while selecting data, or deviations too small to be observed through normal techniques.
A possible solution to some of these issues could be Anomaly Detection (AD), that is the search for data deviating from what is already known. In our field of interest, Particle Physics, we apply this method to look for anomalies (signal data), deviating from well known reacting and decaying particles (background data). The strength of this method lies in its model agnostic nature, meaning that it does not rely on a specific theoretical model to analyse data. This allows us to explore a more vast range of events, increasing the background rejection efficiency, giving us a first hint of the region where to look more in deep for New Physics with precise data analysis techniques. In this process, unsupervised Machine Learning (ML) helps speed up the process.
This thesis investigates whether such an unsupervised approach can be applied to a concrete search. Using ATLAS official Monte Carlo (MC) samples simulating the ATLAS detector during Run-3, a search for BSM resonant signals decaying into hadronic final states with large-radius jets (used to probe boosted resonance topologies) has been conducted, preparing the needed tools to finalise the analysis on real ATLAS data. Leveraging the Anomaly Score (AS) distributions computed by a Graph Neural Network (GNN) for each jet, the analysis strategy was defined, starting with the definition of Working Points (WPs), thresholds on the AS distribution corresponding to a given background rejection fraction. A crucial part of

the strategy was determining the correct modelling function for the background and signals di-jet invariant mass (m_{JJ}) distribution. After tuning and validating a Double-Sided Crystal Ball (DSCB) function to model the signals and a 5-parameter function for the background distribution, fits to the MC samples were executed, deriving the expected upper limits at 95% Confidence Level on m_{JJ} using the CLs method, at different AS WPs. The study on MC samples was concluded and fully validated by performing sensitivity studies based on a binned maximum likelihood fit to the di-jet invariant mass distribution.
The thesis is structured as follows. In Chapter 1 the Standard Model is presented, together with theories of Physics beyond it. The LHC collider and the ATLAS experiment are discussed in Chapter 2. Chapter 3 presents the Anomaly Detection approach. The typical workflow of a Particle Physics analysis is described in Chapter 4. Chapter 5 presents the analysis conducted during this study and its results. Finally, conclusions are drawn in Chapter 6.

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Chapter 1

(Beyond) The Standard Model

The Standard Model (SM) of Particle Physics is a relativistic Quantum Field Theory (QFT) describing three out of four fundamental interactions, strong, weak and electromagnetic (EM), and the behavior of elementray particles under them. For a long time it was attempted to merge the fourth fundamental interaction, gravity, with the SM framework, but those attempts have not been validated yet by experimental observations. Nevertheless, for our purposes, gravity does not play a significant role since the energy scales involved in the processes at a collider experiment, as the LHC, are much lower than the Planck scale, where gravitational effects become relevant. What is remarkable about the SM is that it is a very well tested theory, with predictions confirmed by experimental observations with a very high precision. The discovery of the Higgs boson in 2012 at the LHC, which completed the SM framework, is a clear example of this.
Despite the success of the SM, there are many phenomena that are not explained by it, but are confirmed by astrophysical and cosmological observations, opening the road for Physics Beyond the Standard Model.
This chapter presentes a brief overview of the SM, highlighting its major features. In addition, hints for Physics Beyond it are exposed.

1.1 An overview of the Standard Model

The fundamental constituents of Universe are precisely described by the SM. They are called elementary particles, which are mathematically described as quantum fields $\psi(x)$ and the dynamics is modelled by Lagrangian densities $\mathcal{L}(\psi, \partial_\mu \psi)$ [4]. The SM is a gauge theory, meaning that the Lagrangian is invariant under local transformations of the fields. The gauge group of the SM is

$$SU(3)_C \otimes SU(2)_L \otimes U(1)_Y \tag{1.1}$$

where $SU(3)_C$ describes the strong interaction via Quantum Chromodynamics (QCD), and $SU(2)_L \otimes U(1)_Y$ represents the unified electroweak (EW) interaction.

1.1.1 Particles

1.1.2 Quantum Electrodynamics

1.1.3 Quantum Chromodynamics

1.1.4 Electroweak model

1.2 Hints of Physics Beyond the Standard Model

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The Standard Model (SM) of particle physics is a relativistic Quantum Field Theory (QFT) that provides a comprehensive and extraordinarily precise description of the fundamental constituents of matter and their interactions, excluding gravity [**?**]. The dynamics of these constituents are governed by local gauge symmetries based on the Lie group:

$$SU(3)_C \times SU(2)_L \times U(1)_Y \tag{1.2}$$

where $SU(3)_C$ describes the strong interaction via Quantum Chromodynamics (QCD), and $SU(2)_L \times U(1)_Y$ represents the unified electroweak (EW) interaction, which spontaneously breaks to Quantum Electrodynamics (QED) via the Brout-Englert-Higgs mechanism. This chapter outlines the theoretical framework of the Standard Model and discusses its principal experimental limitations, highlighting the necessity for Physics Beyond the Standard Model (BSM).

1.3 The Standard Model

1.3.1 Particles

The elementary particles described by the Standard Model are divided into two fundamental classes based on their spin quantum number: **fermions** (spin-1/2), which constitute matter, and **bosons** (integer spin), which mediate fundamental interactions and generate mass.

Fermions are further classified into two distinct categories depending on their participation in the strong interaction:

- **Quarks:** Six flavors of quarks exist, organized into three generations of weak doublets:

$$\begin{pmatrix} u \\ d \end{pmatrix}, \quad \begin{pmatrix} c \\ s \end{pmatrix}, \quad \begin{pmatrix} t \\ b \end{pmatrix} \tag{1.3}$$

  Up-type quarks (u, c, t) carry an electric charge of $+2/3\,e$, while down-type quarks (d, s, b) carry $-1/3\,e$. Quarks carry color charge (red, green, blue) and therefore interact via all fundamental forces.

- **Leptons:** Six leptons are similarly grouped into three generations:

$$\begin{pmatrix} \nu_e \\ e^- \end{pmatrix}, \quad \begin{pmatrix} \nu_\mu \\ \mu^- \end{pmatrix}, \quad \begin{pmatrix} \nu_\tau \\ \tau^- \end{pmatrix} \tag{1.4}$$

Charged leptons (e^-, μ^-, τ^-) carry an electric charge of $-1\,e$ and interact via electromagnetic and weak forces. Neutrinos (ν_e, ν_μ, ν_τ) are electrically neutral, massless within the minimal SM, and interact exclusively via the weak force.

Interaction dynamics are mediated by spin-1 **gauge bosons**, which arise naturally from the requirement of local gauge invariance:

- 8 massless **gluons** (g) mediating the strong interaction.
- 1 massless **photon** (γ) mediating the electromagnetic interaction.
- 3 massive vector bosons ($W^\pm, Z^0$) mediating the weak interaction ($m_W \approx 80.38$ GeV, $m_Z \approx 91.19$ GeV).

Finally, a single spin-0 scalar particle, the **Higgs boson** (H, $m_H \approx 125.09$ GeV), completes the particle spectrum, arising as a physical state from the spontaneous breaking of electroweak symmetry [**? ?**].

1.3.2 Quantum Electrodynamics

Quantum Electrodynamics (QED) is the simplest gauge theory within the SM, describing the relativistic quantum field theory of electromagnetic interactions. QED is constructed by requiring local $U(1)_{\text{EM}}$ gauge invariance of the free Dirac Lagrangian for a massive fermion field $\psi(x)$:

$$\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi \tag{1.5}$$

Under a local $U(1)$ phase transformation $\psi(x) \to e^{ie\alpha(x)}\psi(x)$, the derivative transforms non-covariantly. To preserve local gauge invariance, the partial derivative ∂_μ is replaced by the gauge-covariant derivative D_μ:

$$D_\mu = \partial_\mu + ieA_\mu(x) \tag{1.6}$$

where $A_\mu(x)$ is a vector gauge field (the photon field) transforming as $A_\mu(x) \to A_\mu(x) - \partial_\mu\alpha(x)$. Adding the gauge-invariant kinetic term for A_μ yields the full QED Lagrangian:

$$\mathcal{L}_{\text{QED}} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{1.7}$$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor. The interaction term $-e\bar{\psi}\gamma^\mu\psi A_\mu$ dictates the coupling of charged fermions to photons with strength proportional to the charge e.

1.3.3 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the non-Abelian gauge theory describing the strong force acting between color-charged quarks and gluons. QCD is based on the local $SU(3)_C$ gauge symmetry, where C represents the triplet color degree of freedom (red, green, blue). The Lagrangian for N_f quark flavors is given by:

$$\mathcal{L}_{\text{QCD}} = \sum_f \bar{q}_f^a \left(i\gamma^\mu (D_\mu)_{ab} - m_f\delta_{ab}\right) q_f^b - \frac{1}{4}G^A_{\mu\nu}G^{A,\mu\nu} \tag{1.8}$$

where q_f^a represents a quark field of flavor f and color index $a \in \{1,2,3\}$. The gauge-covariant derivative is defined as:

$$(D_\mu)_{ab} = \partial_\mu \delta_{ab} - i g_s T^A_{ab} A^A_\mu \tag{1.9}$$

where g_s is the strong coupling constant, A^A_μ $(A = 1, \ldots, 8)$ are the eight gluon vector fields, and $T^A = \frac{1}{2}\lambda^A$ are the $SU(3)$ generators expressed via the Gell-Mann matrices satisfying $[T^A, T^B] = i f^{ABC} T^C$.

The non-Abelian structure gives rise to the field strength tensor:

$$G^A_{\mu\nu} = \partial_\mu A^A_\nu - \partial_\nu A^A_\mu + g_s f^{ABC} A^B_\mu A^C_\nu \tag{1.10}$$

The presence of the non-linear structure constant term f^{ABC} introduces triple and quadruple gluon self-interactions. This characteristic yields two defining physical properties of QCD:

1. **Asymptotic Freedom:** At high energy scales (short distances), the running coupling constant $\alpha_s(Q^2) = \frac{g_s^2}{4\pi}$ decreases logarithmically, enabling accurate calculations via perturbative QCD (pQCD).

2. **Color Confinement:** At low energy scales (long distances), $\alpha_s(Q^2)$ grows large. Colored particles cannot exist as isolated free states; they undergo fragmentation and hadronization, producing collimated sprays of energetic, neutral hadrons known as **jets**.

1.3.4 Electroweak Model

The unification of electromagnetic and weak interactions is realized through the Glashow-Weinberg-Salam (GWS) model, structured around the local gauge symmetry group $SU(2)_L \times U(1)_Y$. The left-handed fermion components transform as doublets under weak isospin T, whereas right-handed components transform as singlets:

$$\Psi_L = \begin{pmatrix} \nu_L \\ e_L^- \end{pmatrix} \quad (T = 1/2, T_3 = \pm 1/2), \qquad e_R^- \quad (T = 0, T_3 = 0) \tag{1.11}$$

Electric charge Q, weak isospin T_3, and weak hypercharge Y are unified by the Gell-Mann-Nishijima relation:

$$Q = T_3 + \frac{Y}{2} \tag{1.12}$$

The gauge-covariant derivative introducing three W^i_μ $(i = 1, 2, 3)$ bosons for $SU(2)_L$ and one B_μ boson for $U(1)_Y$ is written as:

$$D_\mu = \partial_\mu - i g T^i W^i_\mu - i g' \frac{Y}{2} B_\mu \tag{1.13}$$

Spontaneous Symmetry Breaking (The Higgs Mechanism) Unbroken $SU(2)_L \times U(1)_Y$ symmetry forbids mass terms for gauge bosons and fermions, as explicit mass terms violate local gauge invariance. Mass generation is accommodated via the Brout-Englert-Higgs mechanism by introducing a complex scalar doublet field Φ:

$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \quad \mathcal{V}(\Phi) = \mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 \quad (\mu^2 < 0, \lambda > 0) \tag{1.14}$$

The scalar field acquires a non-zero vacuum expectation value (VEV) $v = \sqrt{-\mu^2/\lambda} \approx 246$ GeV, spontaneously breaking the symmetry:

$$SU(2)_L \times U(1)_Y \xrightarrow{\text{SSB}} U(1)_{\text{EM}} \tag{1.15}$$

Expanding Φ around its minimum in the unitary gauge transforms three degrees of freedom into longitudinal polarization states for the gauge physical bosons $W^\pm$ and Z^0:

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp iW_\mu^2), \quad \begin{pmatrix} Z_\mu \\ A_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix} \tag{1.16}$$

where $\theta_W = \arctan(g'/g)$ is the weak mixing angle. The photon field A_μ remains massless, while gauge bosons acquire masses $m_W = \frac{1}{2}gv$ and $m_Z = \frac{1}{2}v\sqrt{g^2 + g'^2}$. Fermions acquire mass through gauge-invariant Yukawa couplings:

$$\mathcal{L}_{\text{Yukawa}} = -y_f \bar{\Psi}_L \Phi f_R + \text{h.c.} \quad \Longrightarrow \quad m_f = \frac{y_f v}{\sqrt{2}} \tag{1.17}$$

1.4 Hints of Physics Beyond the Standard Model

Despite the extraordinary predictive precision of the Standard Model—culminating in the discovery of the Higgs boson at the LHC in 2012 [**?** **?**]—it cannot be regarded as a complete theory of fundamental interactions. Unresolved theoretical open issues and physical phenomena observed in nature mandate the existence of Physics Beyond the Standard Model (BSM) [**?**]:

- **Dark Matter:** Astrophysical and cosmological observations (galaxy rotation curves, gravitational lensing, cosmic microwave background) firmly establish that non-baryonic *Dark Matter* comprises $\approx 26.8\%$ of the energy density of the Universe [**?**]. The SM offers no suitable stable, cold, weakly-interacting candidate particle [**?**].
- **Neutrino Masses:** Oscillations observed in solar, atmospheric, and reactor neutrinos demonstrate that neutrinos possess non-zero masses, contrasting the strict masslessness assumed in the minimal SM structure.
- **Baryon Asymmetry:** The observable universe consists almost entirely of matter rather than antimatter [**?**]. The magnitude of CP violation provided by the SM CKM matrix is insufficient to account for the observed cosmic matter-antimatter asymmetry [**?**].
- **The Hierarchy Problem:** Quantum loop corrections to the scalar Higgs boson mass quadratic divergences yield corrections proportional to the Planck scale ($\Lambda_{\text{Planck}} \approx 10^{19}$ GeV). Maintaining $m_H \approx 125$ GeV requires extreme, fine-tuned cancellations without a protective symmetry (such as Supersymmetry).
- **Quantum Gravity:** The Standard Model does not incorporate gravitational interactions. A quantum formulation of gravity compatible with General Relativity remains omitted at high energy scales.

These limitations motivate the search for new BSM resonance states and heavy new physics at collider experiments like the Large Hadron Collider (LHC)[cite: 9].

Chapter 2

The LHC and the ATLAS experiments

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A particle physics analysis

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